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SYSTEMATIC REVIEW

Talent Identification in Sport: A Systematic Review

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Abstract

Background Talent identification (TID) programs are an integral part of the selection process for elite-level athletes. While many sport organizations utilize TID programs, there does not seem to be a clear set of variables that consistently predict future success.

Objective This review aims to synthesize longitudinal and retrospective studies examining differences between performance variables in highly skilled and less-skilled athletes in elite-level sport.

Methods The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines were used to identify relevant studies ($N = 20$).

Results There was a clear overrepresentation of studies that (1) examined physical profiles of athletes (60%); (2) focused on male samples (65%); (3) examined athletes between the ages of 10 and 20 years (60%); and (4) were published between the years 2010 and 2015 (65%). On closer examination, there was a high degree of variability in the factors that were found to discriminate between skilled and less-skilled individuals.

Conclusion Findings from this review highlight how little is known about TID in elite sport and emphasize the need for greater diversity in TID research.

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Key Points

This systematic review considers longitudinal and retrospective studies between the years 1990 and 2015 that examined differences in performance variables between highly skilled and less-skilled athletes.

From the studies examined, there was a high degree of variability in the factors that were found to discriminate between skilled and less-skilled individuals. Additionally, there was a clear overrepresentation of studies that examined the physical profiles of male athletes.

This research may be useful for coaches to use in combination with sport-specific and skill-level-specific considerations in determining what level of risk exists for their talent identification decisions.

1 Introduction

Talent identification (TID) programs are designed to identify young athletes with the potential for success in senior elite sport [1]. In recent years, TID programs have grown in popularity and are seen as critical avenues to maximize athletes' potential to achieve success [2, 3]. This is especially true as pressure for nations to excel in sport at the international level is greater than ever [2]. It is not uncommon to see nations investing millions of dollars towards developing evidence-based approaches to finding a competitive edge [4]. This has also been reflected by a surge in research conducted on understanding issues of TID

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and the development of sport expertise over the past 2 decades [5–10].

It has been suggested that an effective TID program has the potential to detect talent early, which may act as a vital component to increasing a nation's chances at sporting success [1]. Anshel and Lidor [2] suggested that TID programs facilitate the athlete selection process by using evidence-based processes that can be refined through feedback and evaluation of the system, thus maximizing the number of gifted individuals at both domestic and international levels. Similarly, Durand-Bush and Salmela [11] noted that TID programs have the capacity to recognize talented athletes early, which helps to focus funding and training opportunities on athletes with the greatest potential for success. However, despite the potential advantages of TID programs, there remains a discrepancy between what is proposed in the research and what is observed in practice [12].

How one perceives talent and ability is important [13], generally reflecting one's perspective on whether exceptional performance is the result of biological or genetically constrained factors (i.e. nature) or the end product of experience and learning (i.e. nurture) [14–17]. While most scientists agree that both factors are important, the nature versus nurture dichotomy continues to dominate popular discourse [18, 19]. Regardless of whether the notion of talent is legitimate or not, misconceptions regarding what talent 'looks like' are widespread in high-performance sport settings. For example, young athletes are often selected on the basis of advantages in physical growth and maturation, as well as the performance benefits conferred by greater size [20]. Yet, neither physical size nor discrete measures of performance at one point in time necessarily equal 'talent' [21, 22].

This lack of understanding regarding the contributions of nature and nurture has led to inconsistencies around the definitions of talent and thus how it might be identified. For instance, Brown [23] described talent as a "special, natural ability" and a "capacity for achievement or success", while Howe and colleagues [17] noted "the likelihood of becoming exceptionally competent in certain fields depends on the presence or absence of inborn attributes variously labeled as talents or gifts" (p. 399). Conversely, Gagné [24] described talent as "the outstanding mastery of systematically developed abilities, called competencies (knowledge and skills), in at least one field of human activity to a degree that places a person at least among the top 10% of age peers who are or have been active in that field" (p. 11). As demonstrated in these examples, there is considerable variation in the definitions, ranging from a focus on innate abilities to outcomes resulting from training and experience.

The very nature of TID in sport is centered on the measurement and subsequent comparison of characteristics that contribute to sport-specific performance. In order to filter out less-talented individuals, researchers often compare different age groups and skill levels in a cross-sectional design [25]. This type of methodology is heavily rooted in assumptions that important characteristics of future success can be extrapolated from individuals' performance at one given point in time [16]. This way of thinking, in its simplest form, implies talent is static as it ignores many important variables such as maturity and relative age effects [20]. However, many of the qualities that distinguish top athletic performance in adults may not be apparent until late adolescence [26, 27], and early performance is not strongly associated with later success [28]. Importantly, because chronological age and biological maturity rarely progress at the same rate, children may be helped or hindered on performance tests by their biological maturity, especially when compared with chronological age norms [29, 30]. Ultimately, these factors reduce the utility of cross-sectional data for informing our understanding of how talent develops and the efficacy of early factors as valid and reliable measures of this outcome. This is in contrast to methodologies that analyze data over a longer period of time (i.e. longitudinal or retrospective designs). These methodologies are often preferred when tracking dynamic variables such as skilled performance as they have a greater capacity to test for causal relationships [31].

Despite the increase in researcher attention to TID and athlete development, evidence regarding the origins of high-level ability has been largely based on cross-sectional designs from unidimensional perspectives. This review aims to (1) gain a better understanding of what is known about this phenomenon by examining longitudinal and retrospective research conducted between 1990 and 2015 examining the differences between performance variables in skilled and less-skilled athletes, and (2) provide evidence-based suggestions to help guide future work in this area.

2 Methods

This review used the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement guidelines [32] to examine the literature on TID in elite sport. A customized search was completed for studies assessing TID in elite-level athletes according to the PRISMA guidelines [32]. Studies were included in the final review if they contained the following:

1. *Highly skilled participants* Only studies examining athletes who fell into the category of 'advanced', 'expert', or 'eminence' were included in this review, as

defined by Baker and colleagues [33]. For example, studies involving physical education in school or 'open-level' sports teams were not included. The purpose of this stipulation was so the focus remained on 'talented' individuals to help understand and monitor the pathway to excellence.

2. *Time-based comparison* The study must have tracked changes in an outcome variable over a period of at least 12 months that allowed comparison between skill groups. To be considered longitudinal, the study must have tracked changes in a performance variable for a minimum of two points over the time period of at least 12 months. To be considered retrospective, at least one performance variable must have been measured at least 12 months before an assessment of success (i.e. world ranking) was conducted. For example, a study including anthropometric and/or skill-based assessments over the course of a week would not meet this criterion as significant changes in performance are unlikely. We wanted to identify studies that measured variables over a reasonable timespan, and, while setting a length as an inclusion criterion is somewhat arbitrary, high-performance coaches are regularly required to make decisions on talent selection over extended time periods. Therefore, we wanted to identify studies that were not short-or medium-term interventions and set 12 months as the minimum study length. This supports the request for systematic studies that track variables for periods longer than days or months [34].
3. *Between-group comparison* Studies must have compared a minimum of two different skilled/talented groups. For example, a study was not included if it examined only athletes competing at the highest level of competition. Without this specification/distinction, it would be difficult to find evidence of factors relating to talent.
4. *Removal of grey-area topics* Studies exploring birthplace effects, deliberate practice, genetic predispositions, handedness, long-term athlete development, or relative age effects were not included in this review. Although relevant for discussions of the notion of talent, each of these topics has a sufficient evidence-base for its own individual PRISMA-based analysis (and in some cases these reviews have already been conducted) [35].

The search strategy for identifying articles was broken down into three phases: phase 1, a search of the electronic databases (in accordance with the PRISMA guidelines [32]); phase 2, a search using additional resources (in accordance with the PRISMA guidelines [32]); and phase 3, a collaboration with a panel of experts. Phase 1 consisted of a search of two electronic databases—Web of Science and SPORTDiscus—in the time period January 1990–July 2015. Studies were identified using the following search terms: 'expertise AND sport', 'talent identification AND sport', and 'giftedness AND sport'. These search terms were found to be commonly used within the literature and were selected to increase the breadth of the search. Phase 2 consisted of a secondary search of external sources such as the reference list of articles found in phase 1, references in books, and additional website searches [36]. The final phase incorporated a panel of three experts who suggested articles that fit the inclusion criteria. After scanning the list of articles from phases 1 and 2, it became evident that the selected researchers were strong contributors to the body of literature on sport expertise. On completion of the three phases, the study's author(s), title, and year of publication were recorded and articles were sorted to eliminate duplicates. From the list of unique entries, the publication's title was read to discern whether the article was written in English and was in the form of a complete, peer-reviewed journal study (i.e. 'reviews', 'commentaries' or 'abstracts' were not included). From this refined list, a more intensive assessment took place, which required obtaining the abstracts and the full-text articles.

3 Results

Phase 1 identified 1696 articles from the database searches using the keywords listed above, with an English-language restriction imposed. An additional 422 articles were identified through external sources, and a final 22 were added from the panel of experts, totaling 444 articles through additional sources. Removal of duplicates resulted in a total of 1695 articles. After reviewing the titles and abstracts, 1296 of these records were eliminated, leaving 399 studies identified for full-text assessment. After a thorough assessment, 379 articles were removed as they did not include a longitudinal or retrospective design, an elite sample of athletes, or a between-group comparison. This left a total of 20 articles that remained in the final study selection (refer to Fig. 1 for a flow diagram of the PRISMA process).

3.1 Descriptive Results

As displayed in Table 1, of the 20 articles included in the review, 16 (80%) of the studies used a longitudinal design, while two studies (10%) used a retrospective design, and the remaining two studies (10%) incorporated a mixed design. From these studies, 19 (95%) were published in the 10-year period between 2005 and 2015 (the remaining study was published in 2004). Furthermore, 60% ($n = 12$) of the articles examined samples between the ages of 10

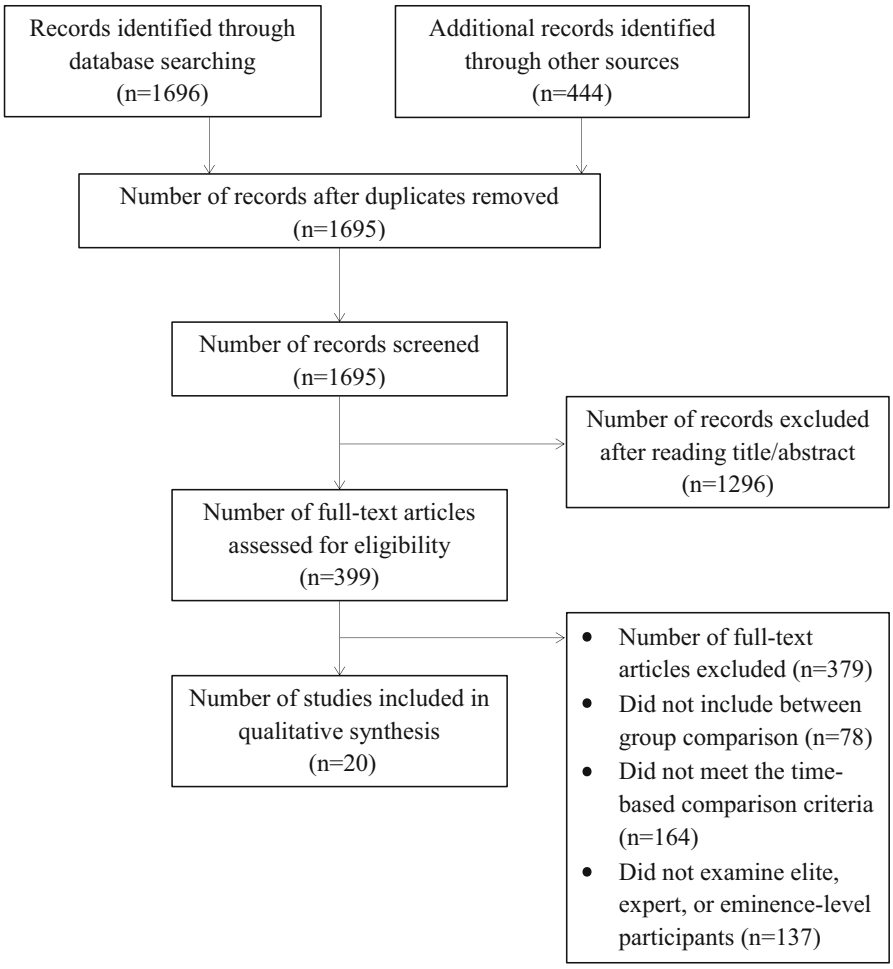

Fig. 1 PRISMA flow chart showing number of records collected and number of eligible records after the screening process. *PRISMA* Preferred Reporting Items for Systematic Reviews and Meta-Analyses

and 20 years, studies with a sample under 10 years of age accounted for 15% ($n = 3$), three studies (15%) included open age categories, one study (5%) had a sample over the age of 20 years, and the remaining study (5%) did not specify the ages of the participants.[1] The majority of the studies ($n = 13$, 65%) examined a male-only sample. Only 10% ($n = 2$) of the studies examined a female-only sample, and the remaining five studies used a combination of both male and female participants. The studies included in this review were nearly all from European countries ($n = 16$), with two studies from Australia and the remaining two studies did not specify the country.

The terminology used by the researchers of the studies to describe the levels of selection varied greatly, with studies using the term elite ($n = 3$), professional ($n = 3$), selected ($n = 3$), drafted ($n = 2$), final selection ($n = 1$), elite cadets ($n = 1$), high division ($n = 1$), national ($n = 1$), phase 3 selected ($n = 1$), senior ($n = 1$), successful ($n = 1$), survivor ($n = 1$), and top world ($n = 1$).

The sport that had the greatest representation was soccer ($n = 7$), followed by gymnastics ($n = 3$) and rugby league ($n = 3$). The remaining studies included Australian Football ($n = 1$), handball ($n = 1$), field hockey ($n = 1$), tennis ($n = 1$), triathlon ($n = 1$), and water polo ($n = 1$). One additional article incorporated multiple sports, including volleyball, swimming, judo, and soccer [28].[2]

The 20 studies included in this review were subdivided into categories according to the types of variables they examined. The first category, cognitive/psychological capabilities and player profiles, included two studies by Van Yperen [38] and Vestberg and colleagues [39]. Van Yperen [38] found that number of siblings, ethnic origin, parental divorce, goal commitment, problem-focused coping, and 'seeking social support' were capable of discriminating between skill levels, while Vestberg and colleagues [39] found that creativity, response inhibition, cognitive flexibility, visual scanning, number sequencing,

[1] Samples in the 10- to 20-year age category likely have a high degree of variability due to the inconsistency of age parameters provided in the studies.

[2] An additional article from Barreiros and Fonseca [37] met our criteria, however due to the similarities to the Barreiros et al. [28] article, the study was not included.

Table 1 Characteristics of studies included in the review

Reference	Sample characteristics						Results
	N	Age, years[a]	Skill level	Sex	Design	Variables examined	
Barreiros et al. [28]	170	U16	Pre-junior	M	Longitudinal	PE	Only one-third of the athletes who were selected to be top junior players were also selected to be on the senior team. This demonstrates the difficulty in using early identification as a predictor for future success
	93	17–18	Junior	M			
	58	19+	Senior	M			
	27	U16	Pre-junior	M			
	21	17–18	Junior	M			
	15	19+	Senior	M			
	60	U16	Pre-junior	M			
	34	17–18	Junior	M			
	18	19+	Senior	M			
	32	U16	Pre-junior	M			
	12	17–19	Junior	M			
	9	20+	Senior	M			
	27	U15	Pre-junior	F			
	15	16–17	Junior	F			
	6	18+	Senior	F			
	64	U14	Pre-junior	F			
	37	15–16	Junior	F			
	21	17+	Senior	F			
	15	U16	Pre-junior	F			
	7	17–19	Junior	F			
	3	20+	Senior	F			
Bottoni et al. [55]	66	14–18	Top world	M	Retrospective	PP	The findings indicate that retrospective analysis of running and swimming performance outcomes is not an appropriate method for predicting future triathlon success
	15	14–18	Top Italian	M			
Brouwers et al. [49]	1897	10–14	Youth	M	Retrospective and longitudinal	PE	Findings demonstrated that player performances at young ages were not correlated with later success in tennis. Additionally, this study did not find an age at which all players should start to perform in order to be successful at the professional level
	281	14	Professional	M			
	68	Open		M			
	202	13–18	Junior	M			
	68	14	Professional	M			
	1624	10–14	Youth	F			
	323	14	Professional	F			
	60	Open		F			
	175	U18	Junior	F			
	60	14	Professional	F			
di Cagno et al. [46]	20	11.5 ± 0.5	Elite cadets	M	Longitudinal	PP	Coordination and precision capabilities can be used as long-term predictors of success in gymnastics
	21	13.3 ± 0.5	Junior cadets	M			
	59	10.5 ± 0.5	Sub-elite cadets	M			
Elferink-Gemser et al. [50]	15	16.0 ± 1.0	Elite	M	Longitudinal	MM	Findings indicated that both female and male elite field hockey players scored better on technical and tactical variables. In addition, female elite-level athletes scored higher on interval endurance capacity and motivation compared with their sub-elite counterparts. Contrastingly, the males in the sub-elite category scored higher in motivation compared with their elite counterparts
	17	16.4 ± 1.3	Sub-elite	M			
	1624	10-14	Youth	F			
	323	14	Professional	F			

Table 1 continued

Reference	Sample characteristics						Results
	N	Age, years[a]	Skill level	Sex	Design	Variables examined	
Falk et al. [51]	60	Open		M	Longitudinal	MM	Selected water polo players were superior on a variety of swimming and motor ability tasks, as well as in game intelligence
	175	U18	Junior	M			
Figueiredo et al. [52]	60	14	Professional	M	Longitudinal	MM	Elite soccer players were found to be older, both chronologically and skeletally, larger, and they out-performed the sub-elite groups in physiological measures and motor skill tests. The degree of goal orientation did not differ between groups
	90	11–14	Club	M			
	33	11–14	Elite	M			
Gil et al. [48]	29	U14	Selected	M	Longitudinal	PP	There were notable differences found among the U14 soccer players who were asked to play in the U15 team compared with the U14 players who were not selected. This selected group of athletes was found to be taller and heavier than their non-selected counterparts
	19	U14	Non-selected	M			
	36	U15	Selected	M			
	17	U15	Non-selected	M			
	29	U16	Selected	M			
	12	U16	Non-selected	M			
	32	U17	Selected	M			
	20	U17	Non-selected	M			
Gil et al. [40]	64	9–10	Pre-selection	M	Longitudinal	PP	The discriminant analysis showed that the selected soccer athletes were older, lighter, and had a lower body mass index rating. The selected individuals also performed better on the velocity and agility tests compared with the control and non-selected groups
	21	9–10	Final selection	M			
	34	9–10	Controls	M			
Gonaus and Müller [44]	205	14	Drafted	M	Longitudinal	PP	Soccer players who were drafted demonstrated superior performance in sport-specific speed and power in the upper limbs as well as other physiological measures compared with non-drafted players
	1160	14	Non-drafted	M			
	252	15	Drafted	M			
	1089	15	Non-drafted	M			
	228	16	Drafted	M			
	995	16	Non-drafted	M			
	136	17	Drafted	M			
	668	17	Non-drafted	M			
Huijgen et al. [53]	53	U12	Selected	M	Longitudinal	MM	Findings indicated that the Loughborough Soccer Passing Test was able to distinguish between players who were selected compared with those who were de-selected
	5	U12	De-selected	M			
	46	U13	Selected	M			
	6	U13	De-selected	M			
	44	U14	Selected	M			
	6	U14	De-selected	M			
	37	U15	Selected	M			
	13	U15	De-selected	M			
	26	U16	Selected	M			
	3	U16	De-selected	M			
	31	U17	Selected	M			
	6	U17	De-selected	M			
	21	U18	Selected	M			
	7	U18	De-selected	M			
	11	U19	Selected	M			
	4	U19	De-selected	M			

Table 1 continued

Reference	Sample characteristics						Results
	N	Age, years[a]	Skill level	Sex	Design	Variables examined	
Lidor et al. [43]	29	12–13	Phase 1 selected	M	Longitudinal	PP	The physiological and anthropometrical tests were not capable of discriminating between the selected and non-selected handball players. The only test that showed a difference between groups was the slalom dribbling test
	118	12–13	Phase 1 non-selected	M			
	24	12–13	Phase 2 selected	M			
	109	12–13	Phase 2 non-selected	M			
	18	12–13	Phase 3 selected	M			
	24	12–13	Phase 3 non-selected	M			
	20	12–13	Phase 1 selected	F			
	54	12–13	Phase 1 non-selected	F			
	20	12–13	Phase 2 selected	F			
	51	12–13	Phase 2 non-selected	F			
	NA	12–13	Phase 3 selected	F			
	NA	12–13	Phase 3 non-selected	F			
Pion et al. [45]	6	6–9	Survivors (continued)	F	Longitudinal	PP	Only 18% of the gymnastics athletes who passed the baseline test consisting of motor skills and physiological measures continued performing at the highest level of competition 5 years later
	85	6–9	Discontinued	F			
Pyne et al. [41]	105	NA	Drafted	M	Longitudinal	PP	Findings showed that the 5 m, 10 m, 20 m sprint, agility test, and the multistage shuttle run discriminated drafted athletes from non-drafted athletes. Of the drafted athletes, those who had better running vertical jump ability and faster agility scores were more likely to debut in an Australian Football League game. Anthropometric measures were not capable of discriminating between drafted versus non-drafted players or debuted players versus non-debuted players
	78	NA	Non-drafted	M			
	166	NA	Debuted	M			
	117	NA	Non-debuted	M			
Till et al. [57]	34	13.6 ± 0.2[b]	Regional	M	Longitudinal	PP	There were significant main effects for selection level, but no significant differences were found for any individual variable to discriminate between selection levels for rugby league players
	19	13.6 ± 0.2	National	M			
	23	13.6 ± 0.2	National-regional	M			
	5	13.6 ± 0.2	Regional-national	M			
Till et al. [56]	249	U15	Amateur	M	Retrospective	PP	Findings illustrated that there were no significant differences between professional and academy rugby league players; however, differences were found between amateur players and professional players
	261	U15	Academy	M			
	70	U15	Professional	M			

Table 1 continued

Reference	Sample characteristics						Results
	N	Age, years[a]	Skill level	Sex	Design	Variables examined	
Till et al. [42]	95	U13	Player performance pathway	M	Longitudinal	PP	Professional U14 and U15 rugby league players outperformed amateur players on the sum of four skinfolds, speed, change of direction, speed and estimated VO_{2max}. Additionally, players who attained professional status were significantly more likely to be later maturing with lower body mass and reduced upper body power compared with amateur and academy players
	50	U13	Amateur	M			
	45	U13	Academy	M			
	13	U13	Professional	M			
	195	U14	Player performance pathway	M			
	92	U14	Amateur	M			
	103	U14	Academy	M			
	18	U14	Professional	M			
	183	U15	Player performance pathway	M			
	107	U15	Amateur	M			
	183	U15	Academy	M			
	39	U15	Professional	M			
Van Yperen [38]	18	16.58 ± 1.4	Successful	M	Longitudinal	CC	There were no differences in levels of recorded exhaustion between the successful and unsuccessful athletes. The successful athletes reported higher engagement in problem-focused coping behavior. The successful athletes were also more likely to seek social support during stressful circumstances and rated their coaches as having a higher performance level. In terms of demographic and other social variables, successful athletes had more siblings, were more often of non-White ethnic origin and were more likely to have divorced parents
	47	16.58 ± 1.4	Unsuccessful	M			
Vandorpe et al. [47]	12	7–8	Elite	F	Longitudinal	PP	In the elite sample of athletes, the non-sport-specific motor test was positively correlated with level of competition; however, the anthropometric and physical characteristics were not
	11	7–8	Sub-elite	F			
Vestberg et al. [39]	14	25.3 ± 4.2	High division	M	Longitudinal and cross-sectional	CC	Soccer athletes in the high division group outperformed their low division counterparts in general executive functioning tasks that are used to demonstrate creativity, response inhibition and cognitive flexibility skills
	17	22.8 ± 4.1	Low division	M			
	15	25.3 ± 4.2	High division	F			
	11	22.8 ± 4.1	Low division	F			

CC cognitive capabilities and player profiles, *F* female, *M* male, MM mixed method, *PE* previous experience/performance, *PP* physical profile, *SD* standard deviation, *U* under, i.e. U20 means under the age of 20 years, *NA* not available, $VO2_{max}$ measure of the maximum volume of oxygen that a person can use

[a]Data are expressed as single year, range, or mean ± SD unless otherwise stated

[b]While the mean age for the total sample of 81 athletes is 13.6 ± 0.2, the mean age for each skill level is not specified

and letter sequencing, were variables with discriminative capabilities.

The second category, physical profile, explored the anthropometric, physiological, and/or sport-specific skills/motor capabilities of the athletes. The majority of studies ($n = 12$) were represented in this category. Some of the variables found to discriminate between the most skilled athletes and the next highest skill level were aerobic capacity [40–42], age/maturation [40], agility [40, 41, 43], height [40], jump height [41], long jump [43], medicine ball throw [43, 44], push up [45], rope jump [45], sit and reach [45], sit up [45], sport-specific drills [43, 45–47], and sprint speed [41, 43–45, 48].

The third category, *previous performance/experience,* examined variables relating to how prior performance and tournament rankings predicted group membership. There

were two studies in this category, by Barreiros and colleagues [28] and Brouwers and colleagues [49]. These variables were not found to discriminate between skill levels.

The remaining four studies (20%) included a combination of the aforementioned categories and were therefore considered mixed measurement studies. Elferink-Gemser and colleagues [50] examined a combination of anthropometric, physiological, sport-specific skill/motor capabilities, tactical skill, and psychological variables. The variables that were found to discriminate skilled females from the less-skilled females were sprint speed, slalom dribble (sport-specific skill), general tactics confidence, and motivation. The skilled male athletes were discriminated by the variables of sprint speed, slalom dribble, general tactics, tactics when in possession, and tactics when not in possession. Falk and colleagues [51] found swimming times and game intelligence to be positively correlated with higher performing athletes, while Figueiredo and colleagues [52] were unable to find any variable capable of discriminating between the highest skill group and the next highest skill group. Lastly, Huijgen and colleagues [53] found the Loughborough Soccer Passing Test (LSPT), a sport-specific soccer drill, capable of discriminating between skill levels.

The inconsistent relationships observed between variables and higher skill levels may be related to a number of different factors such as inconsistent study designs, a lack of research conducted in the field, and a high degree of variability associated with testing unstable variables in discrete ways. Specifically, many of these variables may relate to performance at a given time, however they may not translate to a subsequent level of sport performance later in development. In addition, performance, as well as skill level, is likely constrained by a number of factors related to the task, environment, and the individual [21, 54]. Again, there is a high degree of variability within this type of ecological system, and, as a result, it is unlikely that such discrete and unstable characteristics will consistently demonstrate predictive utility.

4 Discussion

The aim of this study was to review longitudinal and retrospective studies between the years 1990 and 2015, examining the differences in performance variables between highly skilled and less-skilled athletes. Overall, research in TID was mixed, as reflected in the high degree of variability found in the efficacy of different variables in differentiating skill groups (refer to Table 1 for details). While some studies found predictive variables capable of predicting group membership [38, 39, 45, 46, 50], others did not [27, 49, 55, 56]. In general, no variables within the studies examined uniformly predicted skill level. While some variables appeared multiple times in different studies (i.e. height, weight, maturity level, sprint tests, strength tests, and agility tests), no consistent relationship was found between those variables and greater skill. For example, Pyne and colleagues [41] found that anthropometric measures were not capable of discriminating between skill levels in Australian Football, whereas Gil and colleagues [48] found that anthropometric measures did discriminate between selected and non-selected soccer players.

Despite these discrepancies, some variables were consistent across studies. For example, sprint abilities were found to successfully discriminate between skill levels in eight of the studies [41–45, 49, 52, 56]. In addition, agility drills were successfully used to discriminate between skill levels in soccer [40, 52], handball [43], Australian Football [41], and rugby league [42, 56]. Furthermore, three studies examining gymnasts demonstrated successful discrimination between skill levels using the KörperkoordinationsTest für Kinder (KTK) test of coordination and precision in young gymnasts [45–47]. There was also some consistency in variables that did not predict group membership. Barreiros and colleagues [28] and Brouwers and colleagues [49] found no evidence that previous performance/attainment was related to higher skill. Additionally, measures of body composition, as reflected in body mass index (BMI), sum of skin folds, or body fat percentage, were used in a number of studies [40, 41, 45, 47, 48, 52, 57]; however, none was able to distinguish between skill levels of performers.

4.1 Moving Forward: Future Research Directions in Talent Identification

In addition to providing a review of what is a surprisingly limited literature base on TID in elite sport, this review is important for highlighting key areas of future research, specifically in the need for longitudinal research designs. Baker and colleagues [58] draw attention to the advantages of using longitudinal designs to help determine factors influencing skill development. As longitudinal designs help to avoid biases in working with an already talented sample (unlike cross-sectional designs), this may help to decrease the risk of prematurely applying causality to the findings. Despite this advantage, there are some notable drawbacks. Longitudinal designs are time- and energy-intensive and often require large sample sizes to mitigate the high tendencies for dropout and to account for the risk that skilled athletes identified early in life may not continue to be skilled athletes later in life.

It will also be important for future research to consider the terminology of skilled participants. For example, the lack of consistency in the terminology used to express skill levels made it challenging to draw inferences about differences between samples. In this present review, the terms 'elite', 'professional', 'selected', 'national', and 'drafted' were all used to classify skilled participants. It is important to note that even small variations in the way talent is defined may greatly affect how it might be identified, measured, and developed. This finding is not an isolated one as previous research in sporting expertise has highlighted the inconsistencies in the terminology and taxonomy of skill levels [9, 33].

4.2 The Need for More Diverse Research

Perhaps most significantly, the results from the present review highlight the need for a greater diversity in TID research for elite-level athletes. One example can be seen in the significant imbalance between the representation of male (65%), female (10%), and mixed participant (25%) samples. While our general conclusion is that we know very little about predictors of talent in elite sport, we know even less about predicting talent in female athletes. Given the often unique development systems for high-performance female athletes, this discrepancy might limit our ability to gain a deeper understanding of talent, and, as a result, may lead to potentially harmful consequences for the female athlete population. Furthermore, the fact that only three sports (soccer, gymnastics, and rugby league) were represented more than once in this review speaks to how little we know about the vast majority of sports. This lack of diversity makes it very difficult to draw inferences about the predictive utility of testing variables. It also makes it very challenging to isolate a variable that could act as a robust indicator across sport domains.

Another example of how little is known about TID in elite sport can be seen in the number of studies that examined athletes under the age of 10 years ($n = 3$). It is not uncommon to see sport organizations making talent selection decisions early in an athlete's life (as early as childhood), but the lack of research on this age group suggests this practice should be avoided [59]. While early identification might provide some advantages for sport organizations, it may unnecessarily de-select athletes from the athlete development system. Importantly, testing measures used for talent selection are often unstable during prepubertal years. This instability is thought to be a result of the large variation in growth potential in the physical and physiological predictors during the time of testing [24, 25]. In their review, Pearson and colleagues [26] draw attention to the impact that maturity has on testing parameters such as height [60], weight [60], body composition [60–63], anaerobic capacity [64], and strength [65]. While there is some evidence concerning the changes in physical and physiological variables during maturation, we know very little about the stability of cognitive and psychological factors and how they adapt during the early years of an athlete's life. In addition, there have been questions about the stability of these tests given that stress levels, energy levels, and environmental conditions vary during testing, ultimately raising concerns about reliability. It will be important for future research to focus on these considerations and factors in order to enhance the effectiveness of early TID.

It was surprising to see that the majority of the studies in this review were from European nations, with the remaining three studies from Australia. This once again emphasizes the need for more diverse research internationally, particularly in countries such as the US and the UK where TID is an established element of their athlete development programs. Moreover, aside from the study by Vestberg and colleagues [39], no studies in our review explored the predictive utility of perceptual-cognitive skills on the future level of achievement. Given the importance of perceptual-cognitive factors for expert performance [66], this appears to be a potentially fruitful area of future work.

4.3 The Need for a More Ecological Design

Some researchers [67, 68] have suggested that one reason why TID programs are not effective in identifying, selecting, and developing talented athletes is because of the reductionist tendency to deconstruct performance tasks into smaller subphases, which are then used as testing measures in TID programs. This was demonstrated in a number of studies that attempted to isolate the parameters of the sport-specific demands with a simplified agility drill within a sport that is dynamic and interactive, such as soccer [40, 44, 52], handball [43], Australian Football [41], and rugby league [42–46]. As this method does not appear to be effective in representing the demands of competition, it has been suggested that there should be a shift away from this line of thinking in future research [26]. Researchers have also proposed the need for a model that is more representative of performance demands, such as the ecological dynamics model, which places an emphasis on the interactions of the individual in his/her environment where intentions, perceptions, and actions are interconnected rather than treated as separate entities [69, 70]. Although logistically and administratively more complex than univariate approaches, these types of dynamic and multivariate models may better capture the nuances of talent and how it evolves across development. Moreover, given what appears to be limited validity in current approaches, it will

be important to consider more multidimensional and dynamic models when trying to account for interindividual differences in future research.

It has been well-documented that sport is multidimensional in nature, with the optimization of both physical and mental factors being required for elite performance [26, 71]. Despite this, TID research has typically adopted either a unidimensional or restrictive approach by focusing on a select few dimensions of variables (irrespective of known theoretical frameworks) [26, 72, 73]. This approach largely ignores other factors that could influence performance. In particular, there is significant literature highlighting the central role that psychological factors (e.g. coping skills, resilience, confidence, cognitive strategies, determination) play in elite performance [72, 74–79] and how these psychological skills can be incorporated into TID programs [38, 71, 76, 80].

This underrepresentation of multidimensional designs was reflected in the current review, which revealed there was an overrepresentation of studies solely examining the physical profiles of athletes in TID systems. Of the studies selected for inclusion in the review, 60% were focused on physical variables. While many of these studies had test batteries that were quite extensive, they may have been limited by the absence of important psychological and environmental factors [45, 81, 82]. However, four studies embraced the multidimensional approach (in the mixed measurement category) and examined a combination of physical, physiological, and psychological measures. It was likely no coincidence that all four of these studies [50–53] found some predictive utility in the testing variables. This finding echoes the recommendations from Abbott and Collins [72], and speaks to the importance of utilizing a multidimensional approach to allow for more ecologically representative testing designs. Such designs may help increase the chance of finding variables that hold predictive utility for elite-level athletes.

4.4 Limitations of the Review

Although this systematic review provides the first comprehensive synthesis of existing work on predictors of talent in elite sport, it is not without its limitations. One of the main limitations lies in the exclusion of articles that were listed in the 'grey area' (birthplace effect, deliberate practice, genetic predispositions, handedness, long-term athlete development, relative age effect). While the inclusion of these studies would likely enhance the understanding of TID in elite sport, the sheer number of articles would have been too difficult to synthesize in a single review. For example, there were an estimated 34 articles examining relative age effects, along with an additional 30 articles examining deliberate practice, that would have met our criteria. An additional limitation lies in the restriction imposed on articles written in English and published in peer-reviewed journals. It is likely that these restrictions meant we identified only a fraction of the published articles on TID in elite sport. In particular, future reviews may consider incorporating additional keywords (e.g. 'national', 'international', 'intermediate' or 'sub-elite') to expand our understanding of how various outcomes relate to skill acquisition across the process of athlete development.

5 Conclusions

To our knowledge, this is the first systematic review of longitudinal and retrospective studies examining the differences between performance variables in highly skilled and less-skilled participants. The findings of this review have many positive implications for the field of research. Primarily, this review provides a comprehensive synthesis of the past 25 years of research and draws attention to the many gaps in the current body of research. This research may be useful for coaches to use in combination with sport-specific and skill-level-specific considerations in determining what level of risk exists for their TID decisions. This review aimed to synthesize and analyze the past 25 years of research on TID in elite-level sport. Overwhelmingly, findings from this review revealed inconsistent and unreliable predictors and demonstrated a fairly homogenous body of research on TID in elite-level sport. All things considered, it seems reasonable to conclude that there remains a substantial amount of information that we have yet to learn in this field and that future work should include a greater diversity in study designs (e.g. variables, samples, etc.) to reflect the considerable diversity in high-performance sport.

Compliance with Ethical Standards

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References

1. Vaeyens R, Güllich A, Warr CR, et al. Talent identification and promotion programmes of Olympic athletes. J Sports Sci. 2009;27(13):1367–80.
2. Anshel M, Lidor R. Talent detection programs in sport: the questionable use of psychological measures. J Sport Behav. 2012;3(25):239–66.

3. Lidor R, Côté L, Hackfort D. To test or not to test? The use of physical skill tests in talent detection and in early phases of sport development. Int J Sport Exerc Psychol. 2009;7:131–46.
4. Hogan K, Norton K. The 'price' of Olympic gold. J Sci Med Sport. 2000;3(2):203–18.
5. Collings DG, Mellahi K. Strategic talent-management: a review and research agenda. Hum Res Man Rev. 2009;19(4):304–13.
6. Lewis RE, Heckman RJ. Talent-management: a critical review. Hum Res Man Rev. 2006;16(2):139–54.
7. Muller S, Abernethy B, Eid M, et al. Expertise and the spatio-temporal characteristics of anticipatory information pick-up from complex movement patterns. Perception. 2010;39(6):745–60.
8. Nijs S, Gallardo-Gallardo E, Dries N, et al. A multidisciplinary review into the definition, operationalization and measurement of talent. J World Bus. 2014;49(2):180–91.
9. Swann C, Moran A, Piggott D. Defining elite athletes: issues in the study of expert performance in sport psychology. Psychol Sport Exerc. 2014;16:3–14.
10. Williams AM, Ford PR. Expertise and expert performance in sport. Int Rev Sport Exerc Psychol. 2008;1:4–18.
11. Durand-Bush N, Salmela JH. The development of talent in sport. In: Singer RM, Hausenblas HA, Janelle CM, editors. Handbook of sport psychology. 2nd ed. New York: Wiley; 2001.
12. Pankhurst A, Collins D, MacNamara Á. Talent development: linking the stakeholders to the process. J Sports Sci. 2013;31(4):370–80.
13. Wattie N, Baker J. Why conceptualizations of talent matter: implications for skill acquisition and TID. In: Baker J, Cobley S, Schorer J, et al., editors. The Routledge handbook of talent identification and development in sport. London: Routledge; 2017.
14. Baker J, Bagats S, Büsch D, et al. Training differences and selection in a talent identification system. Talent Dev Excel. 2012;4(1):23.
15. Coutinho P, Mesquita I, Fonseca AM, et al. Patterns of sport participation in Portuguese volleyball players according to expertise level and gender. Int J Sports Sci Coach. 2014;9(4):579–92.
16. Davids K, Baker J. Genes, environment, and sport performance: why the nature-nurture dualism is no longer relevant. Sports Med. 2007;37(11):961–80.
17. Howe M, Davidson J, Sloboda J. Innate talents: reality or myth? Behav Brain Sci. 1998;21(3):399–442.
18. Gladwell M. Outliers: the story of success. Hachette: Penguin Books; 2008.
19. Epstein D. The sports gene: inside the science of extraordinary athletic performance. New York: Penguin; 2013.
20. Wattie N, Schorer J, Baker J. The relative age effect in sport: a developmental systems model. Sports Med. 2015;45:83–94.
21. Abbott A, Button C, Pepping GJ, et al. Unnatural selection: talent identification and development in sport. Rev Public Pers Admin. 2005;9:61–88.
22. Wattie N, Cobley S, Baker J. Toward a unified understanding of relative age effects. J Sports Sci. 2008;26:1403–9.
23. Brown J. Sports talent. How to identify and develop outstanding athletes. Champaign: Human Kinetics; 2002.
24. Gagné F. Academic talent development and the equity issue in gifted education. Talent Dev Excel. 2011;3(1):3–22.
25. Breitbach S, Tug S, Simon P. Conventional and genetic talent identification in sports: will recent developments trace talent? Sports Med. 2014;44(11):1489–503.
26. Pearson DT, Naughton GA, Torode M. Predictability of physiological testing and the role of maturation in talent identification for adolescent team sports. J Sci Med Sport. 2006;9(4):277–87.
27. Vaeyens R, Lenoir M, Williams M, et al. Talent identification and development programmes in sport current models and future directions. Sports Med. 2008;35(9):703–14.
28. Barreiros A, Côté J, Fonseca AM. From early to adult sport success: analysing athletes' progression in national squads. Eur J Sport Sci. 2014;14(Suppl 1):S178–82.
29. Malina RM, Coelho e Silva MJ, Figueiredo AJ, et al. Interrelationships among invasive and non-invasive indicators of biological maturation in adolescent male soccer players. J Sports Sci. 2012;30(15):1705–17.
30. Matthys SP, Vaeyens R, Fransen J, et al. A longitudinal study of multidimensional performance characteristics related to physical capacities in youth handball. J Sports Sci. 2013;31(3):325–34.
31. Stritch JM. Minding the time: a critical look at longitudinal design and data analysis in quantitative public management research. Rev Public Person Adm. 2017;37(2):219–44.
32. Moher D, Liberati A, Tetzlaff J, et al. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. Ann Intern Med. 2009;151(4):264–9.
33. Baker J, Wattie N, Schorer J. Defining expertise: a taxonomy for researchers in skill acquisition and expertise. In: Baker J, Farrow D, editors. The Routledge handbook of sport expertise. London: Routledge; 2015. p. 69–79.
34. Abernethy B, Farrow D, Baker J, et al. Developing sport expertise: researchers and coaches put theory into practice. New York: Routledge; 2013.
35. Cobley S, Wattie N, Baker J, et al. A meta-analytical review of relative age effects in sport: the emerging picture. Sports Med. 2009;39(3):235–56.
36. Robertson SJ, Burnett AF, Cochrane J. Tests examining skill outcomes in sport: a systematic review of measurement properties and feasibility. Sports Med. 2014;44(4):501–18.
37. Barreiros AN, Fonseca AM. A retrospective analysis of Portuguese elite athletes' involvement in international competitions. Int J Sports Sci Coach. 2012;7(3):593–600.
38. Van Yperen NW. Why some make it and others do not: identifying psychological factors that predict career success in professional adult soccer. Sport Psychol. 2009;23(3):317–29.
39. Vestberg T, Gustafson R, Maurex L, et al. Executive functions predict the success of top-soccer players. PLoS One. 2012;7:e34731.
40. Gil SM, Zabala-Lili J, Bidaurrazaga-Letona I, et al. Talent identification and selection process of outfield players and goalkeepers in a professional soccer club. J Sports Sci. 2014;32(20):1931–9.
41. Pyne DB, Gardner AS, Sheehan K, et al. Fitness testing and career progression in AFL football. J Sci Med Sport. 2005;8(3):321–32.
42. Till K, Cobley S, Morley D, et al. The influence of age, playing position, anthropometry and fitness on career attainment outcomes in rugby league. J Sports Sci. 2016;4(13):1240–5.
43. Lidor R, Falk B, Arnon M, et al. Measurement of talent in team handball: the questionable use of motor and physical tests. J Strength Cond Res. 2005;19(2):318–25.
44. Gonaus C, Müller E. Using physiological data to predict future career progression in 14- to 17-year-old Austrian soccer academy players. J Sports Sci. 2012;30(15):1673–82.
45. Pion J, Lenoir M, Vandorpe B, et al. Talent in female gymnastics: a survival analysis based upon performance characteristics. Int J Sports Med. 2015;94(11):935–40.
46. di Cagno A, Battaglia C, Fiorilli G, et al. Motor learning as young gymnast's talent indicator. J Sports Sci Med. 2014;13(4):767–73.
47. Vandorpe B, Vandendriessche JB, Vaeyens R, et al. The value of a non-sport-specific motor test battery in predicting performance in young female gymnasts. J Sports Sci. 2012;30:497–505.
48. Gil S, Ruiz F, Irazusta A, et al. Selection of young soccer players in terms of anthropometric and physiological factors. J Sports Med Phys Fitness. 2007;47(1):25–32.

49. Brouwers J, De Bosscher V, Sotiriadou P. An examination of the importance of performance in youth and junior competition as an indicator of later success in tennis. Sport Man Rev. 2012;15(4):461–75.
50. Elferink-Gemser MT, Visscher C, Lemmink KA, et al. Multidimensional performance characteristics and standard of performance in talented youth field hockey players: a longitudinal study. J Sports Sci. 2007;25(4):481–9.
51. Falk R, Lidor R, Lander Y, et al. Talent identification and early development of elite water-polo players: a 2-year follow-up study. J Sports Sci. 2004;22:347–55.
52. Figueiredo AJ, Gonçalves CE, Coelho e Silva MJ, et al. Characteristics of youth soccer players who drop out, persist or move up. J Sports Sci. 2009;27(9):883–91.
53. Huijgen BC, Elferink-Gemser MT, Ali A, et al. Soccer skill development in talented players. Int J Sports Med. 2014;34(08):720–6.
54. Newell KM. Constraints on the development of co-ordination. In: Wade MG, Whiting HTA, editors. Motor development in children: aspects of co-ordination and control. Dodrech: Martinus Nijhoff; 1986.
55. Bottoni A, Gianfelici A, Tamburri R, et al. Talent selection criteria for Olympic distance triathlon. J Hum Sport Exerc. 2011;6(2):294–304.
56. Till K, Cobley S, O-Hara J, et al. Retrospective analysis of anthropometric and fitness characteristics associated with long-term career progression in rugby league. J Sci Med Sport. 2015;18:310–4.
57. Till K, Cobley S, O'Hara J, et al. A longitudinal evaluation of anthropometric and fitness characteristics in junior rugby league players considering playing position and selection level. J Sci Med Sport. 2013;16(5):438–43.
58. Baker J, Hodges NJ, Wilson M. Collecting and assessing practice activity data: concurrent and retrospective approaches. In: Ericsson KA, Hoffman R, Kozbelt A, et al., editors. The Cambridge handbook of expertise and expert performance. 2nd ed. Cambridge: Cambridge University Press **(in Press)**.
59. Bloom BS. Developing talent in young people. New York: Ballantine Books; 1985.
60. Baxter-Jones AD, Helms PJ. Effects of training at young age: a review of the training of young athletes (TOYA) study. Pediatr Exerc Sci. 1996;8:310–27.
61. Roemmich JN, Rogol AD. Physiology of growth and development. Its relationship to performance in the young athlete. Clin Sports Med. 1995;14(3):483–502.
62. Beunen GP, Malina RM, Lefevre J, et al. Prediction of adult stature and noninvasive assessment of biological maturation. Med Sci Sports Exerc. 1997;29(2):225–30.
63. Herman-Giddens ME, Wang L, Koch G. Secondary sexual characteristics in boys: estimates from the National Health and Nutrition Examination Survey III. Arch Pediatr Adolesc Med. 2001;155(9):1022–8.
64. Inbar O, Bar-Or O. Anaerobic characteristics in male children and adolescents. Med Sci Sports Exerc. 1986;18(3):264–9.
65. Hansen L, Bangsbo J, Twisk J, et al. Development of muscle strength in relation to training level and testosterone in young male soccer players. J Appl Physiol. 1999;87(3):1141–7.
66. Mann DT, Williams AM, Ward P, et al. Perceptual-cognitive expertise in sport: a meta-analysis. J Sport Exerc Psychol. 2007;29(4):441–57.
67. Pinder RA, Renshaw I, Davids K. The role of representative design in talent development: a comment on "Talent identification and promotion programmes of Olympic athletes". J Sports Sci. 2013;31(8):803–6.
68. Unnithan V, White J, Georgiou A, et al. Talent identification in youth soccer. J Sports Sci. 2012;30(15):1719–26.
69. Davids K, Araújo D, Vilar L, et al. An ecological dynamics approach to skill acquisition: implications for development of talent in sport. Talent Dev Excel. 2013;5(1):21–34.
70. Pinder RA, Davids KW, Renshaw I, et al. Representative learning design and functionality of research and practice in sport. J Sport Exerc Psychol. 2011;33(1):146–55.
71. MacNamara Á, Button A, Collins D. The role of psychological characteristics in facilitating the pathway to elite performance. Part 2: examining environmental and stage related differences in skills and behaviors. Sport Psychol. 2010;24:74–96.
72. Abbott A, Collins D. Eliminating the dichotomy between theory and practice in talent identification and development: considering the role of sport psychology. J Sport Sci. 2004;22(5):395–408.
73. Hoare D, Warr CR. Talent identification and women's soccer: an Australian experience. J Sports Sci. 2000;18(9):751–8.
74. Durand-Bush N, Salmela JH. The development and maintenance of expert athletic performance: perceptions of world and Olympic champions. J Appl Sport Psychol. 2002;14(3):154–71.
75. Gould D, Maynard I. Psychological preparation for the Olympic Games. J Sports Sci. 2009;27(13):1393–408.
76. MacNamara Á, Button A, Collins D. The role of psychological characteristics in facilitating the pathway to elite performance. Part 1: identifying mental skills and behaviors. Sport Psychol. 2010;24:52–73.
77. MacNamaraÁ Collins D. Profiling, exploiting, and countering psychological characteristics in talent identification and development. Sport Psychol. 2015;29:73–81.
78. Phillips E, Davids KW, Renshaw I, et al. The development of fast bowling experts in Australian cricket. Talent Dev Excel. 2010;2(2):137–48.
79. Phillips E, Davids K, Renshaw I, et al. Expert performance in sport and the dynamics of talent development. Sports Med. 2010;40(4):271–83.
80. Coetzee B, Grobbelaar HW, Grid CC. Sport psychological skills that distinguish successful from less successful soccer teams. J Hum Mov Stud. 2006;51(6):383–402.
81. Baker J, Horton S. A review of primary and secondary influences on sport expertise. High Abil Stud. 2004;15(2):211–28.
82. Côté J. The influence of the family in the development of talent in sport. Sport Psychol. 1999;13(4):395–417.